

Name:	Class:	Class Register Number:
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中正中學

CHUNG CHENG HIGH SCHOOL (MAIN)

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PRELIMINARY EXAMINATION 2026 SECONDARY 4

ADDITIONAL MATHEMATICS

4049/01

Paper 1

Tuesday 25 August 2026

2 hours 15 minutes

Candidates answer on the Question Paper.

READ THESE INSTRUCTIONS FIRST

Write your name, class and index number in the spaces at the top of this page.
Write in dark blue or black pen.
You may use an HB pencil for any diagrams or graphs.
Do not use paper clips, glue or correction fluid.

Answer **all** the questions.
Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place in the case of angles in degrees, unless a different level of accuracy is specified in the question.
The use of an approved scientific calculator is expected, where appropriate.
You are reminded of the need for clear presentation in your answers.

At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [] at the end of each question or part question.

The total number of marks for this paper is 90.

For Examiner's Use	
Question Number	Marks Obtained
1	
2	
3	
4	
5	
6	
7	
8	
9	
10	
11	
12	
Total Marks	

This document consists of **16** printed pages.

[Turn over

Mathematical Formulae**1. ALGEBRA***Quadratic Equation*

For the equation $ax^2 + bx + c = 0$,

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Binomial expansion

$$(a + b)^n = a^n + \binom{n}{1} a^{n-1} b + \binom{n}{2} a^{n-2} b^2 + \dots + \binom{n}{r} a^{n-r} b^r + \dots + b^n,$$

where n is a positive integer and $\binom{n}{r} = \frac{n!}{r!(n-r)!} = \frac{n(n-1) \dots (n-r+1)}{r!}.$

2. TRIGONOMETRY*Identities*

$$\sin^2 A + \cos^2 A = 1$$

$$\sec^2 A = 1 + \tan^2 A$$

$$\operatorname{cosec}^2 A = 1 + \cot^2 A$$

$$\sin(A \pm B) = \sin A \cos B \pm \cos A \sin B$$

$$\cos(A \pm B) = \cos A \cos B \mp \sin A \sin B$$

$$\tan(A \pm B) = \frac{\tan A \pm \tan B}{1 \mp \tan A \tan B}$$

$$\sin 2A = 2 \sin A \cos A$$

$$\cos 2A = \cos^2 A - \sin^2 A = 2 \cos^2 A - 1 = 1 - 2 \sin^2 A$$

$$\tan 2A = \frac{2 \tan A}{1 - \tan^2 A}$$

Formulae for $\triangle ABC$

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

$$a^2 = b^2 + c^2 - 2bc \cos A$$

$$\Delta = \frac{1}{2} ab \sin C$$

- 1 In triangle ABC , $AB = (2 + \sqrt{3})$ cm and $\angle BAC = 60^\circ$. The area of the triangle is $(6 + 3\sqrt{3})$ cm². Without using a calculator, find the length of AC , expressing your answer in the form $k\sqrt{3}$ cm, where k is an integer. [4]

- 2 The points A and B lie on the curve $xy = 8$. The midpoint of AB is $M(3,3)$. Find the coordinates of A and of B . [5]

- 3 (a) Sketch the graph of $y = 2 \cos x$ for $0^\circ \leq x \leq 360^\circ$. [2]

- (b) By adding a suitable straight line to your sketch, find the number of solutions of the equation $4 \cos x + 1 = 0$ for $0^\circ \leq x \leq 360^\circ$. [3]

4 A curve has equation $y = e^{2x} + 3$ and it meets the y -axis at the point P .

(a) Find the equation of the tangent to the curve at P .

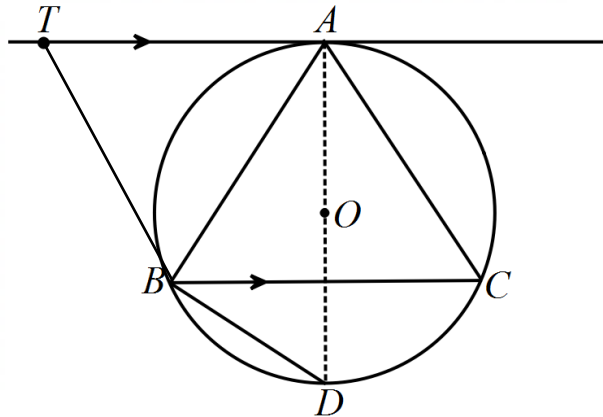
[4]

The equation of the normal to the curve at P meets the x -axis at Q .

(b) Find the coordinates of Q .

[3]

- 5 In the diagram below, TA and TB are tangents to a circle, centre O , at the points A and B respectively. The chord BC is parallel to TA and AD is the diameter of the circle.



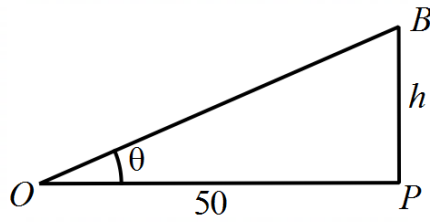
- (a) Prove that triangle ABC is isosceles.

[3]

- (b) Prove that triangle TAO is congruent to triangle TBO . Hence, state the name of the quadrilateral $TAOB$.

[4]

- 6 The diagram shows a balloon at B , rising vertically from a fixed point P on level ground. An observer at O is 50 m from P , and the angle of elevation of the balloon from O is θ radians. The balloon rises at a constant rate of 4 m/s.



At the instant when the balloon is 30 m above the ground, find the rate at which

- (a) the angle of elevation θ is increasing,

[5]

(b) the distance of the balloon from the observer is increasing.

[4]

- 7 Given that $\cos A = \frac{3}{5}$ and $\sin B = \frac{5}{13}$, where A and B are acute and $A > B$, find the exact value of

(a) $\sin 2A$, [2]

(b) $\cos(A + B)$, [2]

(c) By writing $A - B = 2A - (A + B)$ and using your results to parts (a) and (b), find the exact value of $\sec(A - B)$. [4]

8 A bank offers two saving schemes for a small annual interest rate x , invested over n years.

- **Scheme A:** Amount invested multiplies by $(1+x)^n$,
- **Scheme B:** Amount invested multiplies by $(1+2x)^6$.

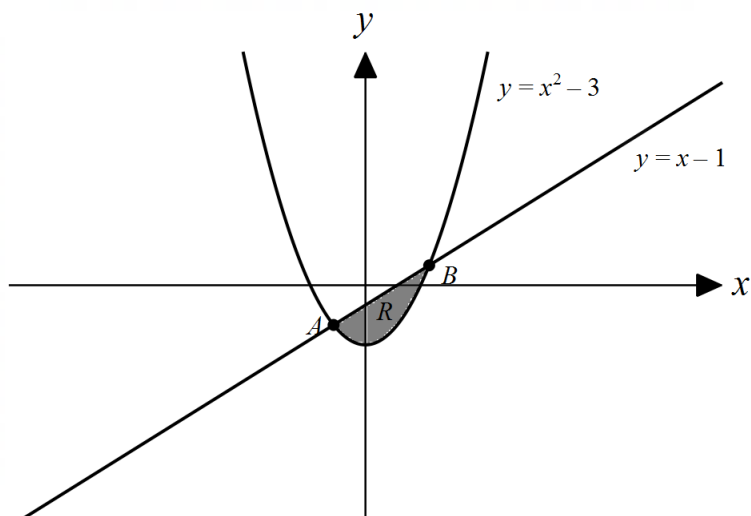
(a) The bank designs Scheme A so that in the expansion of $(1+x)^n$, the coefficient of x^2 is 66.
Show that $n=12$. [2]

(b) Write down the expansions of $(1+x)^{12}$ and $(1+2x)^6$, each in ascending powers of x up to and including the term in x^2 . [2]

(c) Hence, determine which scheme gives the greater return for small positive values of x , justifying your answer. [2]

(d) Given that $x=0.02$, estimate the difference between the two returns and hence the difference in the final amount for an initial investment of \$20 000. [2]

- 9 The diagram shows the curve $y = x^2 - 3$ intersecting the line $y = x - 1$ at the points A and B . The shaded region R is enclosed by the curve and the line.



- (a) Find the coordinates of A and of B .

[4]

- (b) Find the area of the shaded region R .

[4]

- (c) For a value of x in the interval between the x -coordinates of A and B , the vertical distance between the line and the curve is denoted by p . Express p in the form $a - (x - b)^2$, and hence state the maximum value of p . [5]

10 (a) Express $\frac{9x+9}{(x-1)(x+2)^2}$ in partial fractions.

[4]

(b) Hence, find $\int \frac{9x+9}{(x-1)(x+2)^2} dx$.

[2]

11 (a) Express $8\sin\theta + 6\cos\theta$ in the form $R\sin(\theta + \alpha)$, where $R > 0$ and $0^\circ < \alpha < 90^\circ$. [3]

(b) Hence, solve the equation $8\sin\theta + 6\cos\theta = 5$ for $0^\circ \leq \theta \leq 360^\circ$. [4]

(c) State the maximum value of $4 + 8\sin\theta + 6\cos\theta$ and the value of θ , where $0^\circ \leq \theta \leq 360^\circ$, at which the maximum occurs. [2]

12 A cuboid has a square base of side x cm and height h cm. The total length of all twelve edges is 36 cm.

(a) Show that the volume of the cuboid, $V \text{ cm}^3$, is given by $V = x^2(9 - 2x)$. [2]

(b) Explain why the value of x must satisfy $0 < x < 4.5$. [2]

(c) Given that x can vary, find the maximum volume of the cuboid and the value of x at which this occurs. Justify that this volume is a maximum. [5]